

Chapter 1

Introduction

Memory protection is a fundamental security mechanism in modern processors, ensuring that software components cannot access memory regions that are not authorized to use. In RISC-V systems, the **Physical Memory Protection (PMP)** extension provides a mechanism to enforce memory protection, allowing Machine-mode code to be protected from User- and Supervisor-mode access. However, if Machine-mode code is sufficiently complex, vulnerabilities can arise and without protections within Machine-mode this becomes a problem. This is why **Enhanced Physical Memory Protection (smePMP)** was developed: it enables additional restrictions on Machine-mode while also keeping the capabilities of **PMP**.

To ensure these security implications are met, formal verification can be used rather than relying solely on testing. Formal verification provides mathematical proofs for statements you make. Using these mathematical proofs we can formally formulate both the security guarantees that are expected and the definition of the extension—in this case **PMP**. Once these are both defined we can prove whether the model meets the specified security guarantees.

Katamaran [1] is a tool that provides a framework to make this easier and is already applied to verify **PMP**. It does however currently not include the **smePMP** extension. Extending Katamaran’s **PMP** definition to include **smePMP** is necessary to formally verify the security guarantees. However, this extension is non-trivial, we must first add the status register required by **smePMP** before updating the extension’s definition. Once the definition is updated, the formal proofs must also be extended to reflect **smePMP**’s security guarantees.

The contribution of this thesis is to extend the Katamaran prover’s RISC-V **PMP** case study to formally verify the **smePMP** extension. This involved three main steps: first, understanding the existing **PMP** verification in the Katamaran codebase, second, adding the **mseccfg** register with the correct fields required by **smePMP** and third, updating the prover’s **smePMP** definition and the associated proofs.